Model Memory vs Retrieval

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

WORKBOOK · 1 OF 21

Recall and choose your starting point

> Record your starting explanation for Model Memory vs Retrieval before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Verify Numbers and Dates

Without opening the old lesson, explain one key idea from Verify Numbers and Dates. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Hallucination Failure Types

Without opening the old lesson, explain one key idea from Hallucination Failure Types. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Visual Composition

Without opening the old lesson, explain one key idea from Visual Composition. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Model Memory vs Retrieval trace and record the first step you can explain without help.

Use W2: Trace the worked case for **Retrieval** and **index**ing. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Model Memory vs Retrieval explanation against the supplied facts before discussing it.

Use the worked case for **Retrieval** and **index**ing. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Apply a new case independently

Corpus has public P, teacher-only T and withdrawn W. A student search returns all three. Identify the filtering defect and whether hiding T after generation is sufficient.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Model Memory vs Retrieval target to a trace and a justified conclusion.

A model states the library opens at 8 from learned patterns. A supplied current notice says 9. Explain the stronger answer and one remaining check.

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Model Memory vs Retrieval case and show the deciding evidence.

One assistant answers without search; another retrieves a handbook paragraph.

Compare **evidence** trails.

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

WORKBOOK · 7 OF 21

Check a changed case

Apply the procedure to the changed case and record why the conclusion follows.

A model recalls last year's opening hours; a retrieved current notice gives different hours. How should the answer use this **evidence**?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 17 Test and reflect

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

Project checkpoint

> Justify what Model Memory vs Retrieval adds to Verified Study Buddy and keep the evidence.

The Study Buddy retrieves course passages and never presents model memory as a citation.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

A model states the library opens at 8 from learned patterns. A supplied current notice says 9. Explain the stronger answer and one remaining check.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 21

W1 Explain the mechanism

> Explain the mechanism. Apply the supplied Model Memory vs Retrieval facts and keep your reasoning.

Where to find the supplied example

Reference for this exercise: open this week's student lesson, "Model Memory vs Retrieval", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

WORKBOOK · 13 OF 21

W2 Trace the worked case

> Trace the worked case. Apply the supplied Model Memory vs Retrieval facts and keep your

> reasoning.

Index v1 contains notice N1 for room A. **Corpus** v2 supersedes it with N2 for room B, but the query still returns N1. What can be reported? Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

WORKBOOK · 14 OF 21

W3 Audit the limit

> Audit the limit. Apply the supplied Model Memory vs Retrieval facts and keep your reasoning.

A report omits this qualification: High **retrieval** similarity does not override permissions or make an obsolete source current. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Apply a new case independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply a new case independently. Apply the supplied Model Memory vs Retrieval facts and keep your reasoning.

Corpus has public P, teacher-only T and withdrawn W. A student search returns all three. Identify the filtering defect and whether hiding T after generation is sufficient.

My reasoning and evidence

W5 Design a revealing follow-up

> Design a revealing follow-up. Apply the supplied Model Memory vs Retrieval facts and keep your reasoning.

Where to find the supplied example

Reference for this exercise: open this week's student lesson, "Model Memory vs Retrieval", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

WORKBOOK · 17 OF 21

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 18 OF 21

Pictorial case practice: Model Memory vs Retrieval

> The library assistant remembers an old opening time, but a current notice can be retrieved. Explain the labelled step before checking the changed condition.

The library assistant remembers an old opening time, but a current notice can be retrieved.

Approved notice LIB-v = 2 says Wednesday closes at 16: 00; the model previously produced 15: 00. The question asks the current Wednesday closing time.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-3/17

Model Memory vs Retrieval

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Old guess

A remembered-looking answer says 15: 00; no current notice is attached.

2. Find notice

Retrieve authorised LIB-v = 2, which explicitly says Wednesday closes at 16: 00.

3. Ground answer

Answer 16: 00 and attach LIB-v = 2. Its words, rather than the earlier guess, support this claim.

CHANGE ONE THING

LIB-v = 2 is unavailable.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why can the assistant not certify 15: 00 when the notice is missing? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Verified Study Buddy record.

Project: Verified Study Buddy

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

The library assistant remembers an old opening time, but a current notice can be retrieved.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Answer 16: 00 and attach LIB-v = 2. Its words, rather than the earlier guess, support this claim.

1. Old guess

2. Find notice

3. Ground answer

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Ground answer Answer 16: 00 and attach LIB-v = 2. Its words, rather than the earlier guess, support this claim. Read the picture in order.

Answer 16: 00 and attach LIB-v = 2. Its words, rather than the earlier guess, support this claim. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: LIB-v = 2 is unavailable. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. This one example guarantees correct results for every future **input**.

3. Report that the current time cannot be verified.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Report that the current time cannot be verified. The unsupported old guess does not become current **evidence** when **retrieval** fails.

Repair a missing building block

Review Verify Numbers and Dates (ai-3/16). Goal: Explain the **mechanism** and terminology of claim support and citation evaluation.

Short practice: Which part can be corrected immediately from existing **evidence**, and which needs follow-up?. Return to this week and explain what you changed.

Review Hallucination Failure Types (ai-3/14). Goal: Explain the **mechanism** and terminology of generation failures and verification. Short practice: Can we conclude there will definitely be no lunch from S alone?.

Return to this week and explain what you changed.

Review Visual Composition (ai-3/11). Goal: Explain the **mechanism** and terminology of image generation and provenance. Short practice: Which checks pass after that single change, and what still needs repair?.

Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

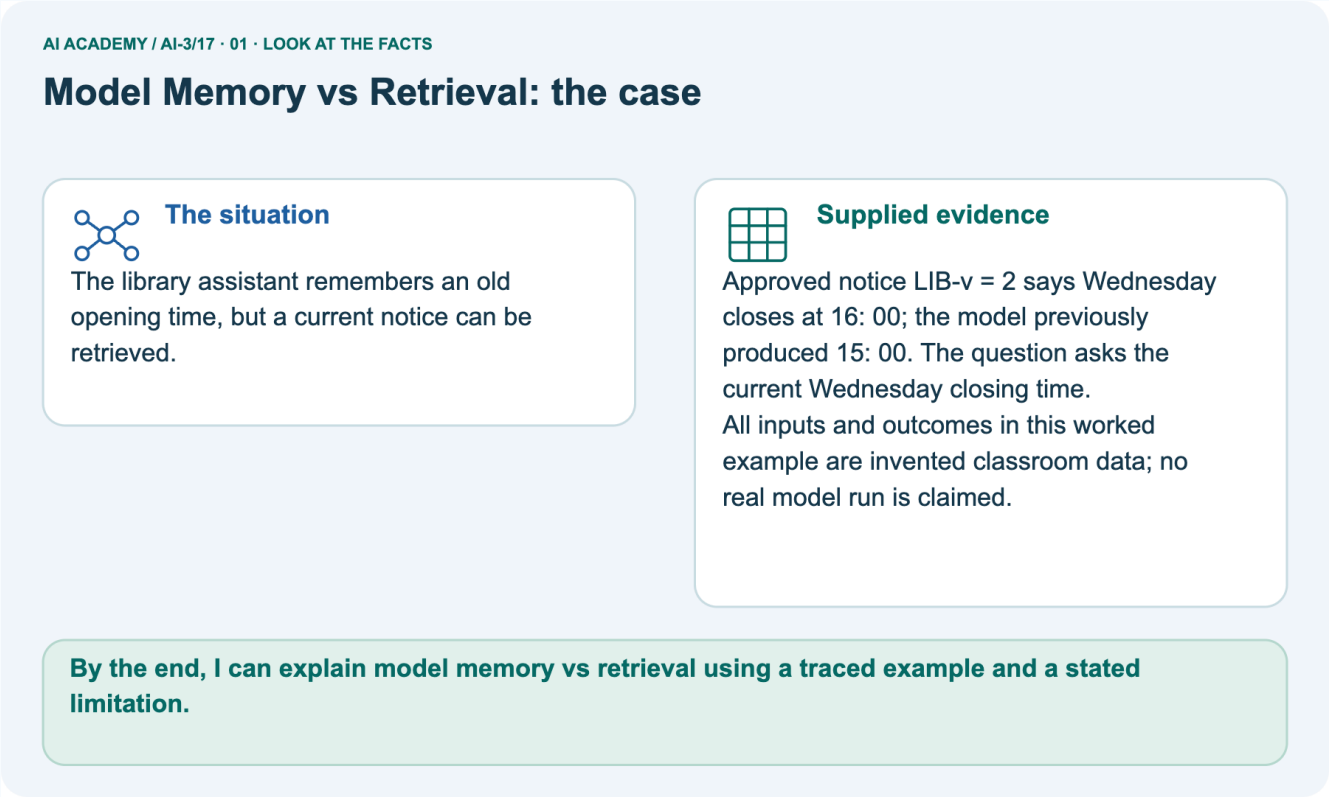
Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

Visual concept journey

> Explain each Model Memory vs Retrieval picture from its labels, then record a relevant limit.

01. Model Memory vs Retrieval: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

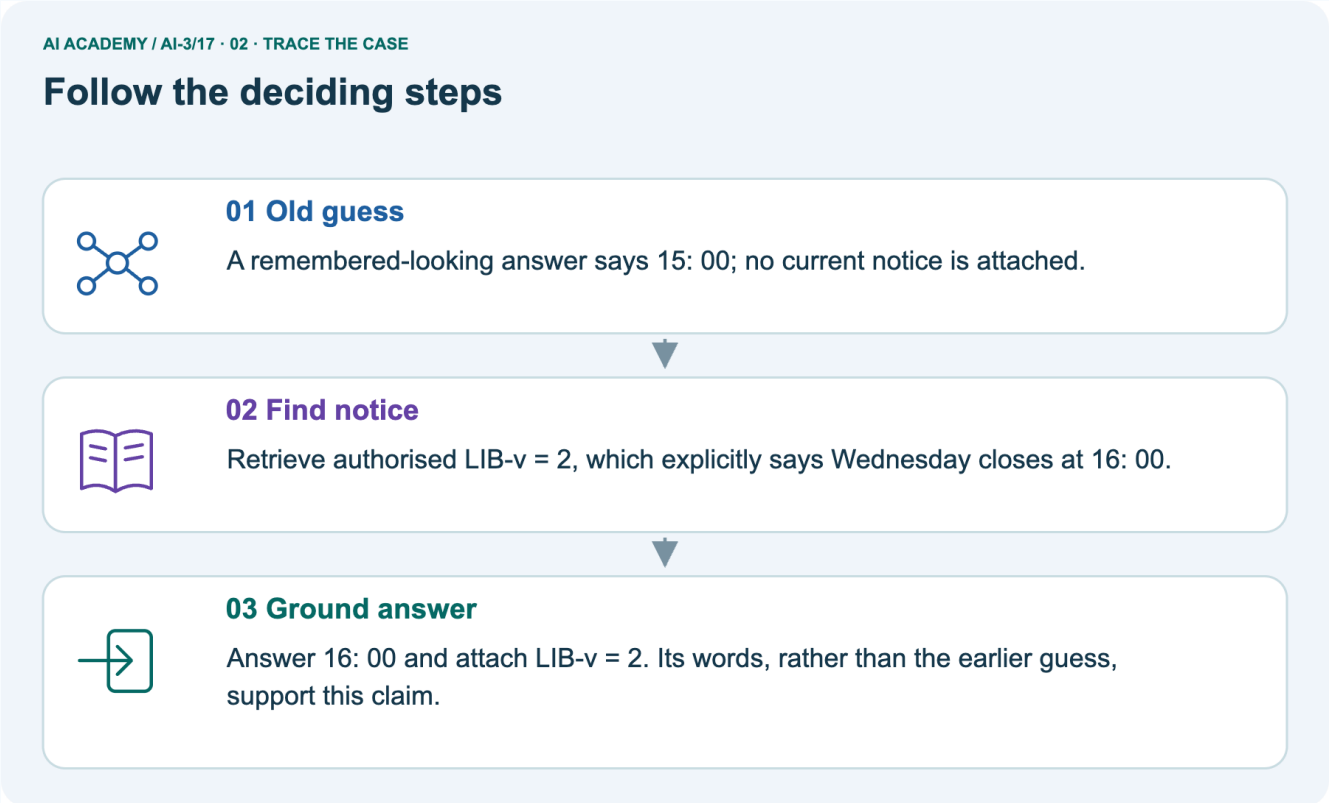
The library assistant remembers an old opening time, but a current notice can be retrieved. Supplied facts: Approved notice LIB-v = 2 says Wednesday closes at 16: 00; the model previously produced 15: 00.

The question asks the current Wednesday closing time. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

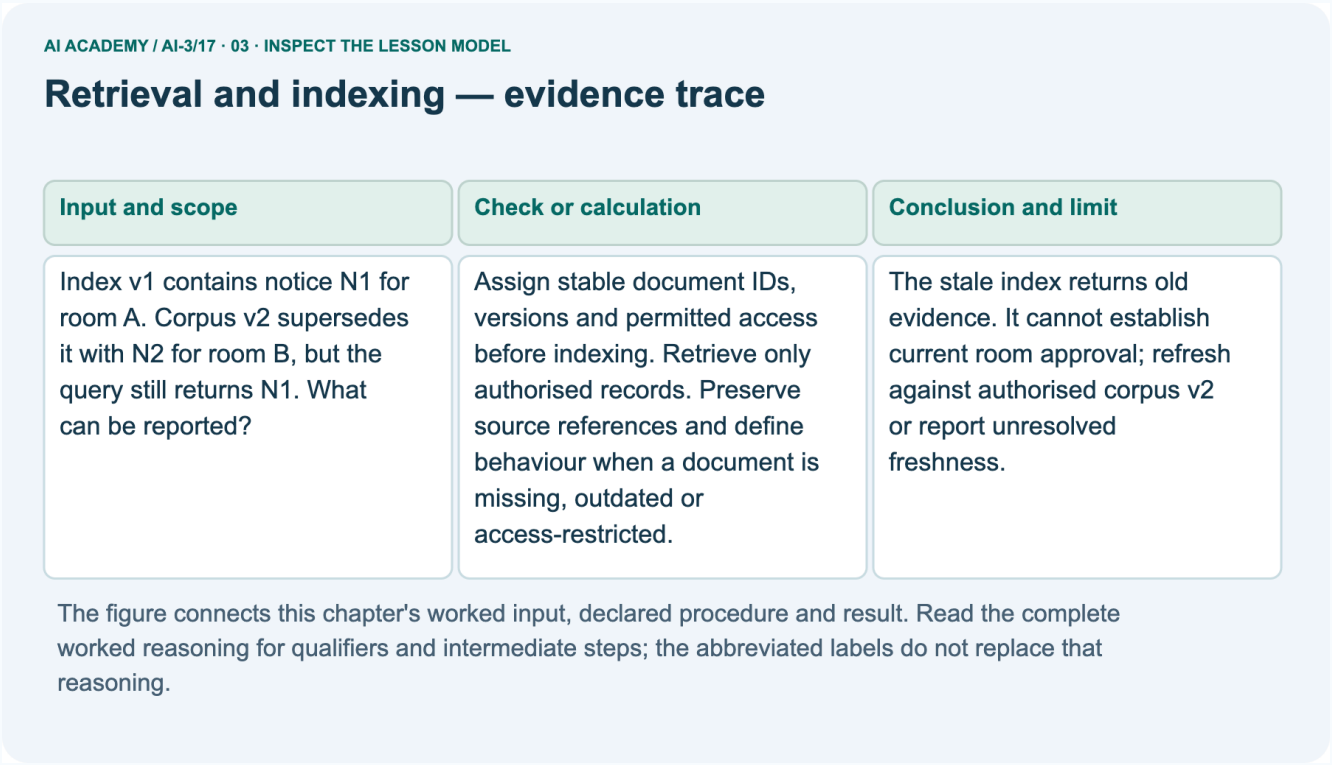
Old guess: A remembered-looking answer says 15: 00; no current notice is attached. → Find notice: Retrieve authorised LIB-v = 2, which explicitly says Wednesday closes at 16: 00. → Ground answer: Answer 16: 00 and attach LIB-v = 2.

Its words, rather than the earlier guess, support this claim.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Retrieval and indexing — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Retrieval and **index**ing — **evidence** trace. **Index** v1 contains notice N1 for room A. **Corpus** v2 supersedes it with N2 for room B, but the query still returns N1.

What can be reported? | Assign stable document IDs, versions and permitted access before **index**ing. Retrieve only authorised records.

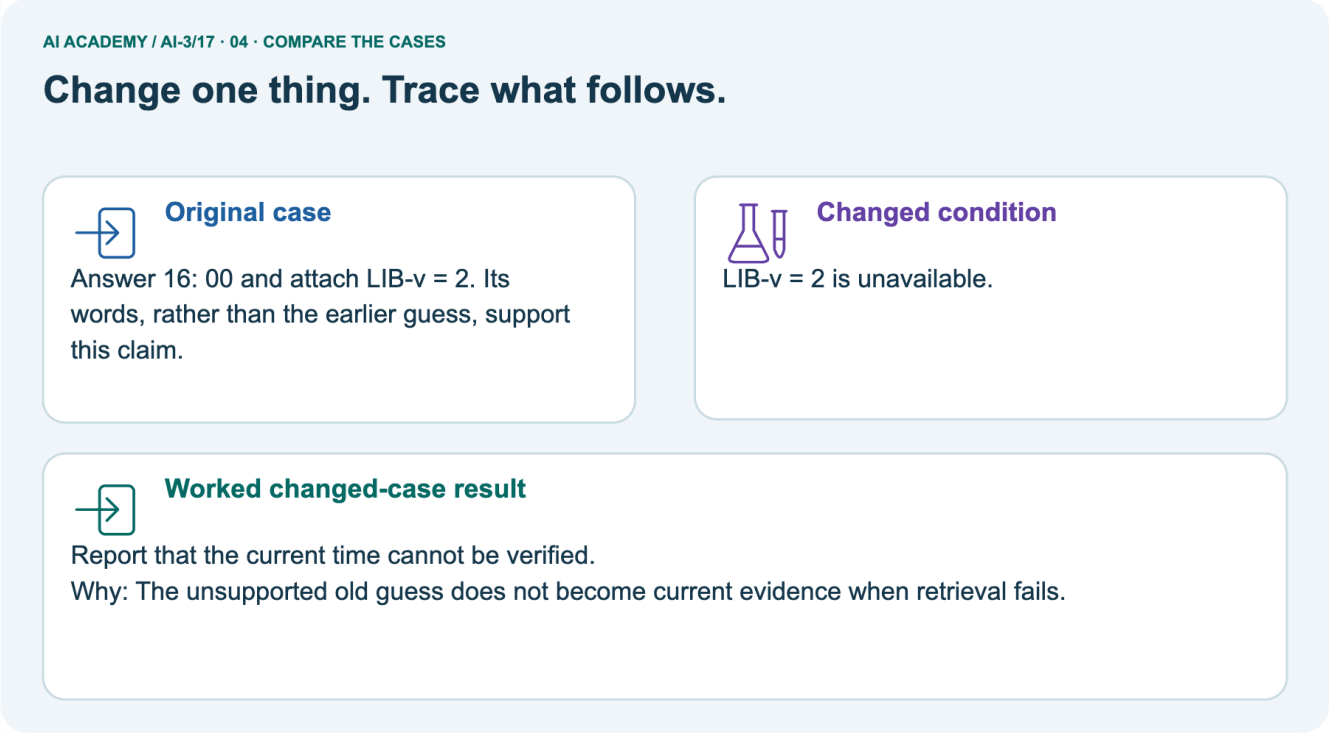
Preserve source references and define behaviour when a document is missing, outdated or access-restricted. | The stale **index** returns old **evidence**.

It cannot establish current room approval; refresh against authorised **corpus** v2 or report unresolved **freshness**.

> Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

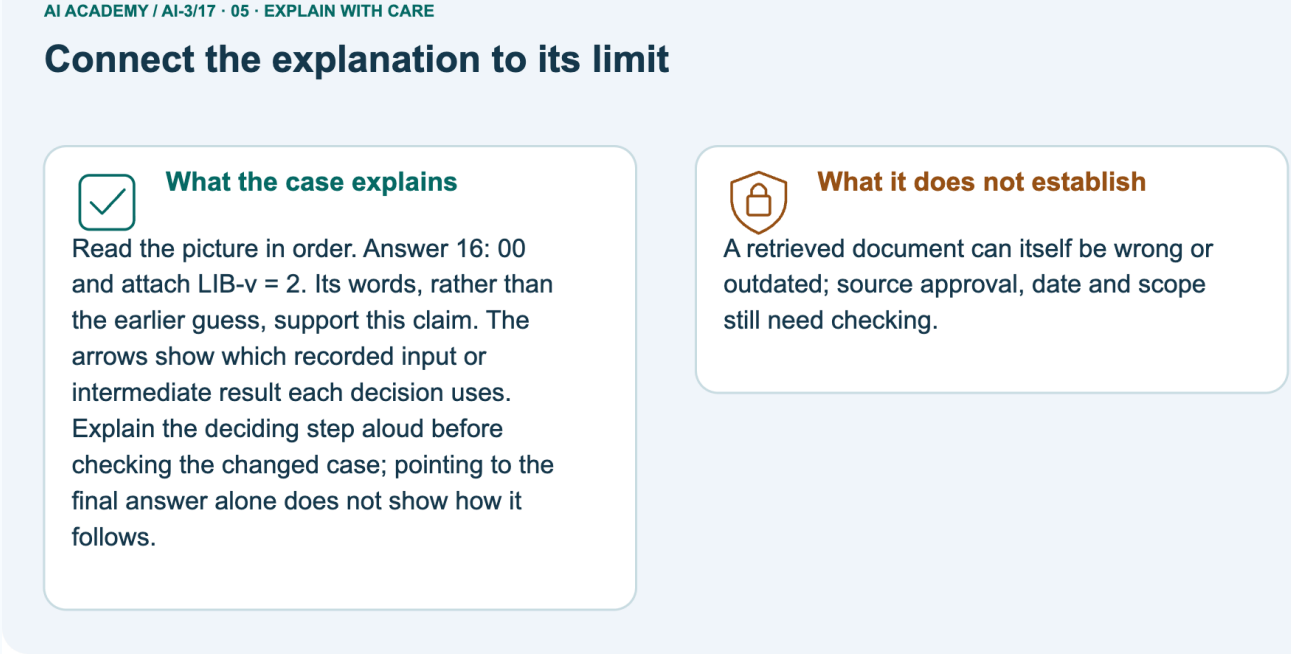
Original case: Answer 16: 00 and attach LIB-v = 2. Its words, rather than the earlier guess, support this claim. Changed condition: LIB-v = 2 is unavailable. Expected result: Report that the current time cannot be verified.

Reason: The unsupported old guess does not become current **evidence** when **retrieval** fails.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Answer 16: 00 and attach LIB-v = 2. Its words, rather than the earlier guess, support this claim. The arrows show which recorded **input** or intermediate result each decision uses.

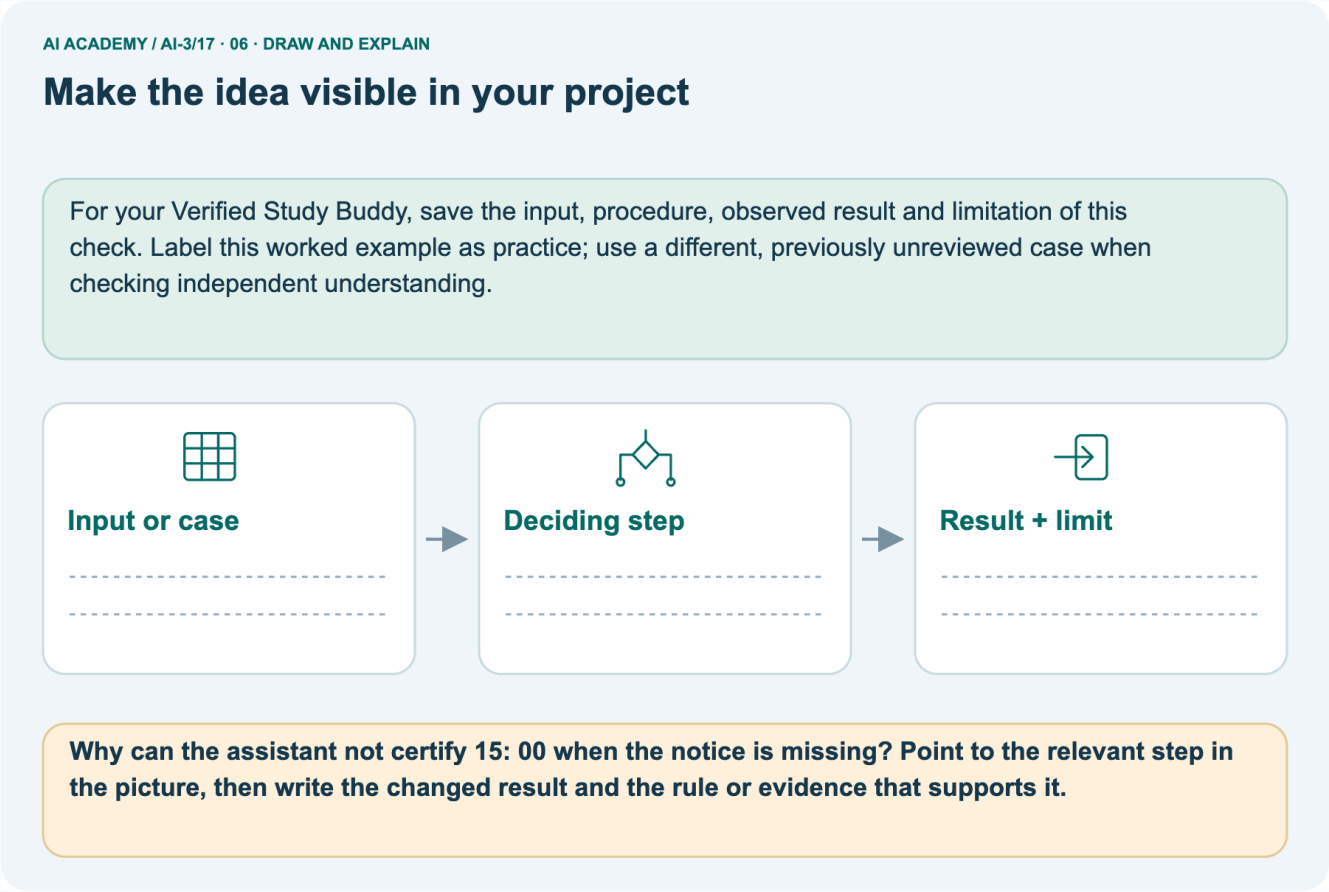
Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: A retrieved document can itself be wrong or outdated; source approval, date and scope still need checking.

> Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why can the assistant not certify 15: 00 when the notice is missing? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.